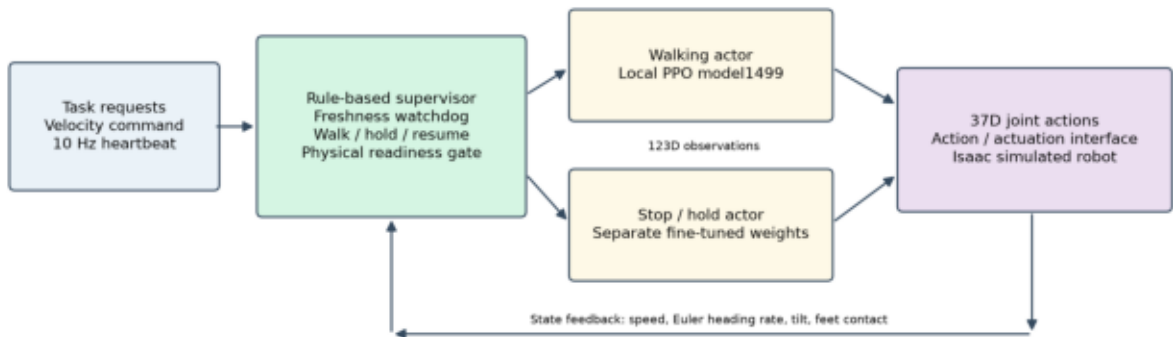


Hierarchical G1 Control in Simulation

Skills, handoffs and watchdogs | 4 October 2026 (Sydney)

Research question: Can separately trained walking and stop/hold actors be coordinated through explicit interfaces, state-based handoff conditions and a task-message watchdog?

Implemented: a rule-based supervisor above two PPO actors, with compatible 123 D observations and 37 D joint actions. The walking actor was trained locally from random weights; the hold actor was separately fine-tuned from that foundation. This is hierarchical control, not a learned HRL manager.



Held-out replication: fixed policies and thresholds

Requested task	Pass / cases
Normal walk / hold / walk	108 / 108
Temporary 1s task-message loss: stop / resume	108 / 108
Persistent loss: bounded 12s hold-actor observation	108 / 108

324 new cases: reset seeds 101/202/303, onset bases 3.22/6.22/9.22 s, four staggered copies, yaw commands -0.2/0/+0.2 rad/s. All cases retained; previous seed 42 results not pooled. Seed variation changes initial world root pose, not joints or velocities. Copies within a seed are not independent trials.

Measured medians: stop confirmation 1.84 s after the stop decision; request-to-stop net displacement 8.83 cm(normal) /20.64 cm(loss, including watchdog delay); hold net drift about 0.9 cm; resume tracking confirmation 1.72 s. Detection delay 0.24-0.30 s. Persistent observation includes stopping: confirmed hold median 10.16 s.

Evidence checks: 318,455 first-episode action rows reconstructed independently, 216 matched prefixes, maximum actor error 3.34 e-6; observation, action history, gating, watchdog, metrics and explicit horizons verified.

Scope: flat Isaac at 0.5 m/s without perturbations or observation noise. Task-heartbeat loss is simulated, not a perception dropout. MuJoCo stationary transfer, high-speed/rough/disturbance tests, fallen recovery and hardware safety remain open.

Sources: claims.csv, evidence_chain.md and frozen_demo.json in this brief folder.

Research decisions and next questions

Reproduce -> diagnose -> train for handoff states -> integrate -> retest

1. Build the locomotion foundation

Reference workflow: 4096 environments, 1500 PPO updates, 147,456,000 transitions on an 8 GB RTX 4060 Laptop. Local baseline trained from random weights; reference robot/task/rewards/PPO settings attributed.

2. Reject a misleading improvement

500 additional updates improved some Isaac metrics but candidate 1999 terminated 9/12 MuJoCo conditions. It was not promoted. Cross-simulator directional control remains a separate limitation.

3. Separate diagnostics from control authority

Passive monitor trajectory equivalence was verified; detection calibration did not meet all requirements. Zero walking commands and fixed postures were tested before assuming a stop/recovery capability.

4. Investigate skill entry states

684 static/direct/braking cases showed that handoff conditions influence survival. Some apparent post-handoff passes disappeared when braking displacement was included. Longer braking was not uniformly better.

5. Train on the incoming state distribution

A 15,872-state bank from walking 1499 supported 500 additional updates for each target actor. Hold improved under selected Isaac conditions; marching showed rhythmic motion but failed original full in-place criteria.

6. Integrate and retain negative ablations

The two-actor walk/hold/walk sequence passed 48/48 with direct switching, 43/48 with action blending and 37/48 with braking plus blending. Direct switching was selected before held-out replication.

7. Freeze a bounded experimental demonstration

Three new reset seeds and expanded entry times yielded 108/108 passes in each of three task/fault scenarios. Raw traces, protocol/source/weight hashes, phase metrics and independent verification are retained.

Research relevance and next questions

Research on hierarchical interfaces, skill sequencing, watchdog handling, timing analysis and quantitative simulation-first experimentation. Proposed next research: connect genuine perception/task interfaces, profile compute jitter, broaden incoming-state and disturbance coverage, and calibrate recovery before supervised hardware validation.

Reproducibility: compact verified tables/configuration are bundled; full raw traces, training logs and weights remain local and Git-ignored. The labeled demo displays Isaac recorded poses on MuJoCo geometry, with zero physics steps. Early baseline provenance gaps are disclosed; later hashes cannot repair training-time snapshots.

Reference foundation: https://github.com/yezzzzye/g1_walk_isaacclab_mujoco

No new locomotion algorithm, general robustness or hardware safety claim.